

Concurrent synchronization and covariation in upper extremity joint angles during a reciprocal aiming task

Marijn S.J. Hafkamp^{1,2,3,*}, Tim A. Valk², Raoul M. Bongers²

¹ Department of Computational Cognitive Science, Tilburg School of Humanities and Digital Sciences, Tilburg University, Warandelaan 2, 5037 AB, Tilburg, The Netherlands

² Department of Human Movement Sciences, University Medical Center Groningen, University of Groningen, Antonius Deusinglaan 1, 9713 AV, Groningen, The Netherlands

³ Institut des Sciences du Mouvement, Aix-Marseille Université, CNRS, 163 Avenue de Luminy, 13009, Marseille, France

* M.hafkamp@tilburguniversity.edu

ABSTRACT

The ecological-dynamical approach holds that movement is self-organized, implying a circular causality between the kinematics of the end-effector and the coordination of degrees of freedom (DOF) into synergies. Yet, how this circular causality manifests itself in redundant effector systems remains underexplored, because proper models for multi-DOF systems are lacking. Therefore, we investigated concurrently the coordination dynamics of the end-effector and the upper extremity joint angles –as DOF– during a reciprocal aiming task with varying target widths and distances. Results revealed that different joint angles were differently related to the end-effector. A continuous relative phase analysis showed that the three joint angles with the largest range of motion synchronized with the end-effector, suggesting a strong dynamical coupling between the levels. At the same time, all joint angles exhibited variability that was not shared with the end-effector. A detrended fluctuation analysis revealed signatures of fractality as well as drift in this residual variability at the DOF level, suggesting distinct dynamical regimes. Moreover, the drift observed in the joint angles occurred predominantly along the uncontrolled manifold, leaving the end-effector unaffected. We propose that the DOF of a redundant system are not only coordinated into synergies to stabilize the end-effector, but also to keep the system flexible.

Introduction

In human movement, the many available degrees of freedom (DOF, e.g., muscles, joint angles) of the movement apparatus are typically organized into stable coordinative structures [1-3]. Within the ecological-dynamical approach, these structures are known as *synergies*: temporary assemblies of DOF that are constrained to act as functional units [4-7]. Such synergies are understood to be self-organizing, meaning that the kinematic outcome of the movement and the underlying coordination of the DOF –respectively the macro- and microscopic level of organization– are linked through circular causality. When aiming to a target, for instance, there is reciprocity between the kinematics of the end-effector and the coordination of the joint angles in the upper extremity [8]. In this reciprocity, the end-effector trajectory emerges from the coordination of the joint angles, while the realm of possible coordination patterns in the joint angles is simultaneously constrained by the trajectory of the end-effector. In model systems with only a few kinematic degrees of freedom (c.q. [13]), this process of self-organization is relatively well understood. Yet, how circular causality

manifests itself in redundant effector systems such as the arm and hand –which has more DOF than necessary to aim for a target– remains underexplored [9,10]. Proper models of circular causality in multi-DOF systems are to date unavailable. The present study therefore investigated concurrently the coordination dynamics of the upper extremity joint angles and the end-effector during a reciprocal aiming task [11,12], to explore the process of self-organization in a redundant movement system.

A seminal account of self-organization in movement is provided by the Haken–Kelso–Bunz (HKB) model of rhythmic interlimb coordination [13]. In this model, two rhythmically moving limbs are treated as a pair of self-sustaining oscillators with a non-linear coupling function. Their coordination can be captured macroscopically by a potential function with only two attractor states: in-phase (0°) and anti-phase (180°). At low movement frequencies, both attractors are stable and relative phase variability is minimal. As the movement frequency increases, however, the anti-phase attractor loses stability. This leads the system to switch to the more stable in-phase coordination [14,15]. Thus, the state of the attractor depends on the rhythmic frequency of the oscillators, while the oscillators' rhythmicity is constrained by the attractor state. This reciprocity illustrates the model's circular causality between the macro- and microscopic level of organization.

Soon after its introduction, the coupled-oscillator model became a general operational paradigm for studying self-organization in movement [16-18]. By integrating system modeling with empirical work, the paradigm uncovered some of the core principles of coordination dynamics. Notably, it demonstrated that the stability of a macroscopic attractor depends not only on the two oscillator's frequencies, but also on the symmetry between the rhythmic components. When the oscillators differ in their eigenfrequency, the faster oscillator takes the lead and the stability of their (anti-phase) coordination declines immediately [19,20]. Similarly, when two oscillators are coupled asymmetrically, their synchrony is disrupted and their coordinative stability decreases as well [21]. The fundamental insight gained from these models is that coordinative stability on a macroscopic level hinges upon the capacity of system components on a microscopic level (i.e., DOF) to establish stable phase relationships. Consequently, synergy came to be associated with the synchronization of DOF [7,9,17], here defined as an in-phase/anti-phase coupling.

However, for tasks in which an end-effector needs to be positioned in space, synchronization is not the only route to coordinative stability. Because the movement system is inherently redundant (or abundant [22]) for most tasks, multiple configurations of the system's DOF can produce the same outcome at the macroscopic level. Therefore, stability at this level can also emerge from mutual compensation (i.e., covariation) between the system's DOF [3,22,23]. For instance, the same hand movement can be brought about by extending the elbow while keeping the shoulder fixed, or by rotating the shoulder while holding the elbow steady. In goal-directed movement, this covariation in joint angles ensures that the target can be reached in multiple ways using different patterns of coordination, which is fundamental to the system's flexibility. The idea of motor abundance has been linked to the concept of the uncontrolled manifold (UCM) [23,24]. The UCM is a subset of DOF configurations (represented as a manifold in the system's state space) that yield the same movement outcome. Think, for instance, of all joint angle combinations that keep the effector on a target. In line with this

idea, synergy has been re-interpreted [3] as a grouping of DOF that covaries within an uncontrolled manifold, as a way to stabilize a movement while preserving its flexibility.

In the ecological-dynamical approach, the two models of synergy –HKB and UCM– are often used interchangeably [10]. Synchronization and covariation have even been described as complementary features of one and the same synergy concept. They would each contribute to coordinative stability in their own way [5]. For goal-directed movement, however, the models yield different predictions about the system’s coordination dynamics. Coupled oscillator models, on the one hand, suggest that the DOF of a redundant system must be synchronized for the end-effector trajectory to be stable. Studies on the coordination dynamics of reciprocal aiming have shown that the end-effector can behave as a self-sustaining oscillator, suggesting a limit cycle attractor at the macroscopic level [12,25]. To preserve the stability of this attractor, its dynamics must somehow be reflected in the back-and-forth movement of the joint angles at the microscopic level. This dynamical reflection ultimately implies synchronicity as a basis for the joint angle coordination.

The UCM model, on the other hand, suggests a weaker synchronization between the macro- and microscopic level of system organization. Studies of discrete [26,27] and reciprocal [8] aiming have shown that the joint angles of the upper extremity covary to stabilize the trajectory of the end-effector. This suggests that the angles are free to deviate from the end-effector dynamics, because they can (co)vary along an uncontrolled manifold. At the same time, there must be *some* synchronization between the levels, for the joint angles ultimately drive the end-effector motion. Taking the HKB and UCM approaches together, there seems to be an ambiguous relationship between the macro- and the microscopic level of organization in redundant effector systems.

Using a cross-recurrent quantification analysis (CRQA), Valk and colleagues [8] explored this ambiguity in the context of a continuous aiming paradigm. CRQA is a non-linear analysis that quantifies how patterns of motion in one time series (e.g., in the end-effector) recur in another time series (e.g. in a joint angle) at different temporal offsets, without specifying the dynamics of either signal. They found that only three out of the nine upper extremity joint angles –the shoulder plane of elevation (SPE), the shoulder exo-endorotation (SEE) and the elbow flexion-extension (EFE)– had a strong dynamical coupling to the end-effector, while six other angles –the shoulder elevation (SEL), elbow pronation-supination (EPS), wrist flexion-extension (WFE), wrist abduction-adduction (WAA), finger flexion-extension (FFE) and the finger abduction-adduction (FAA)– were only weakly coupled (see **Table 1** for all joint angles). This suggests that not all joint angles synchronise similarly with the end-effector. Moreover, it suggests that some angles display dynamical regimes that are not observed at the end-effector level. Yet, neither a CRQA nor a UCM captures a system’s dynamics, leaving unresolved how the two levels of organization dynamically relate to one another during aiming.

Table 1: joint angle abbreviations

Abbreviation	Joint angle
SPE	Shoulder plane of elevation angle
SEE	Shoulder exorotation-endorotation angle

SEL	Shoulder elevation angle
EFE	Elbow flexion-extension angle
EPS	Elbow pronation-supination angle
WFE	Wrist flexion-extension angle
WAA	Wrist abduction-adduction angle
FFE	Finger flexion-extension angle
FAA	Finger abduction-adduction angle

Therefore, the current study examined the coordination dynamics of the upper extremity joint angles and the end-effector during a reciprocal aiming paradigm [11,12]. In this task, participants made back-and-forth aiming movements between two targets, while target distance and target width were both manipulated. First, we investigated the degree of synchronization between the joint angles and the end-effector. Following up on the work of Valk et al. [8], we conducted a relative phase analysis and hypothesized that only the three ‘coupled’ joint angles – the SPE, SEE and EFE – would be locked into in-phase or anti-phase relationships with the end-effector. Second, we investigated the deviations from synchrony in the joint angle movements, the so-called ‘residual variability’. Since we had no a priori hypotheses regarding the dynamical regimes at the joint angle level, we conducted our analyses in an exploratory fashion, focusing on the temporal evolution of the considered DOF.

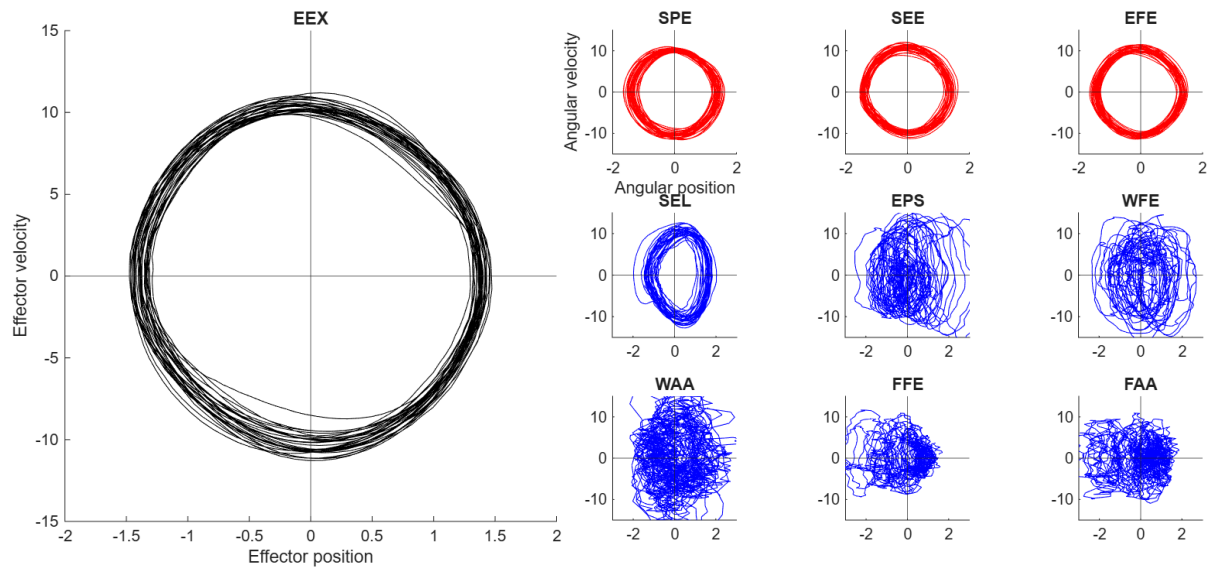
Results

Originally, this study was designed to compare aiming conditions of varying target distance (5, 10, 20 and 30 cm) and target width (index of difficulty 3.5 - 6.0 with increments of 0.5). Our aim here, however, was to compare the coordination dynamics of the joint angles with the end-effector. Accordingly, our analyses focused on *within-trial* effects. For comparisons between aiming trial conditions, we refer the reader to Valk et al. [8].

Synchronicity between joint angles and end-effector

Phase plane portraits

In **Figure 1** we reconstructed the normalized phase plane portraits of the end-effector (**1A**) and the joint angles (**1B**) of the upper extremity for a representative participant in an exemplary condition (index of difficulty 5.0, distance 20 cm). As illustrated in these portraits, three of the nine joint angles (SPE, SEE and EFE, in red) displayed limit-cycle dynamics that were comparable to that of the end-effector (established by continuous relative phase analysis, see below). The six remaining joint angles (SEL, EPS, WFE, WAA, FFE, FAA, in blue) were much more variable in their dynamics, some of them exhibiting no apparent structure.



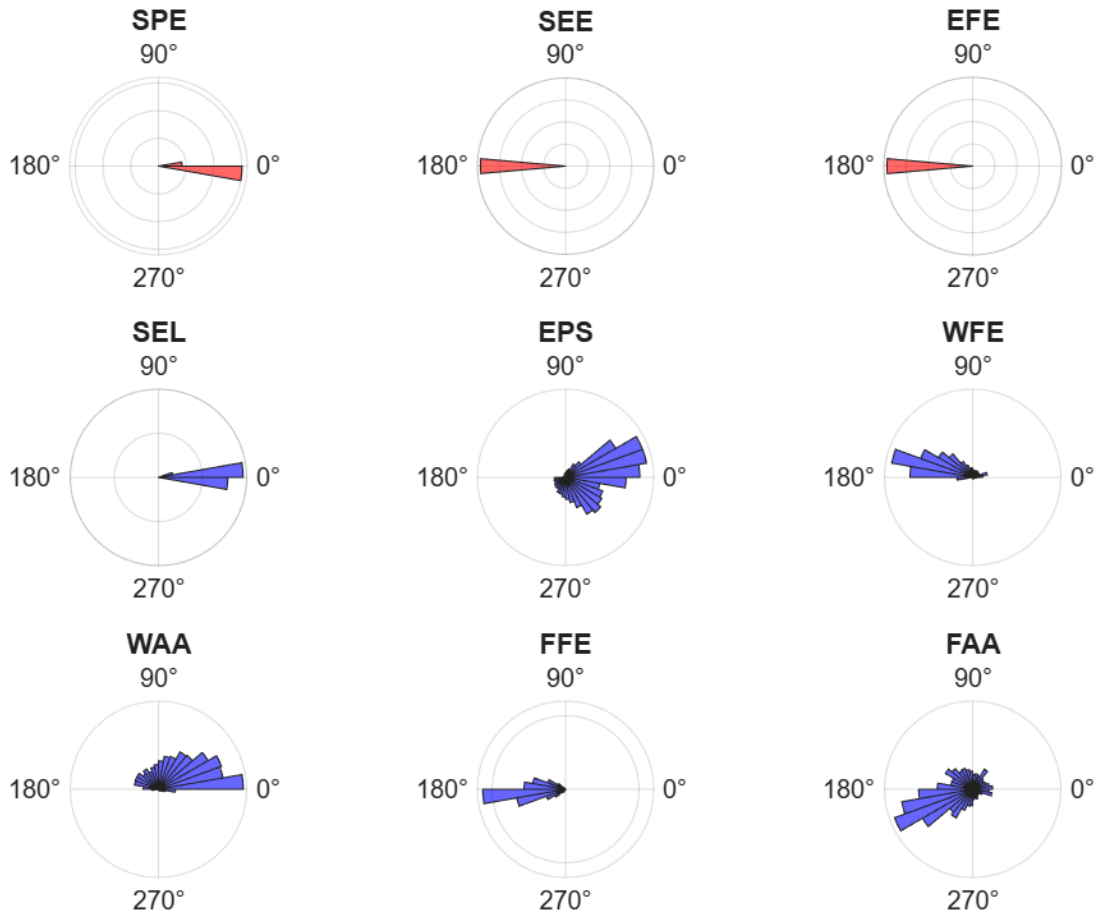
159 **Figure 1**

160 ***Continuous relative phase analysis***

161 To examine the synchronization between the joint angles and the end-effector, we computed
 162 the continuous relative phase (CRP) [28] with tightened normalization procedures [29]
 163 between the corresponding time series. In **Figure 2** we showed the distribution of the CRP for
 164 each joint angle, for the same participant and trial condition as in **Fig. 1**. The polar histograms
 165 reveal that the CRP fluctuated around 0° or 180° for most angles, most notably for the SPE,
 166 SEE and EFE angles. This suggests high degrees of synchronization with the end-effector. At
 167 the same time, the CRP fluctuated strongly for angles such as the wrist abduction-adduction
 168 (WAA, between 0° or 180°) and the finger abduction-adduction (FAA, between 90° and 270°).
 169 This suggested lower degrees of synchronization.

170

171



172 **Figure 2**

173 To assess the degree of synchronization for each joint angle, we computed the phase locking
 174 value (PLV) for all CRP time series across participants and conditions. The PLV was defined
 175 such that in-phase and anti-phase synchronization would both correspond to a coefficient of
 176 1, while a random distribution of CRP values (i.e., no synchronization) corresponded to a
 177 coefficient of 0. As shown in **Figure 3**, the PLV of the SPE, SEE and EFE (in red) was close to 1
 178 in all conditions, indicating a very high degree of synchronization. In the other six angles (in
 179 blue), the PLV was substantially lower, ranging from 0.5 to 1.0. This demonstrated a weaker
 180 synchronization between these joint angles and the end-effector, pointing at deviations from
 181 the macroscopic dynamics. Following analyses further investigated these deviations.



Figure 3

Deviations from synchronicity

Residual variability in the joint angles

To further assess the deviations of joint angles from macroscopic dynamics, we separated the variability in the joint angles that *deviated* from the end-effector dynamics from that which was shared with the end-effector. To do this, we conducted a first order regression analysis on the normalized angular time series of each joint angle, with the normalized end-effector trajectory as a predictor. This resulted in residual variability time series per trial, which we illustrated in **Figure 4** for the same participant and condition as presented in **Fig. 1** and **2**. The residual variability corresponds to the variation in the joint angle timeseries that is *not* shared with the end-effector, because the linear model was removed from the data (the linear model was not of interest for our current purpose). As can be observed, this variability was small for the joint angles that strongly synchronized with the end-effector (SPE, SEE and EFE, residual variability depicted in red), but much larger for the other joint angles (in blue). In the latter

case, it displayed (1) distinct spatiotemporal structures and (2) a strong extent of relative drift in the joint angle over time. The next two sections systematically study both phenomena.

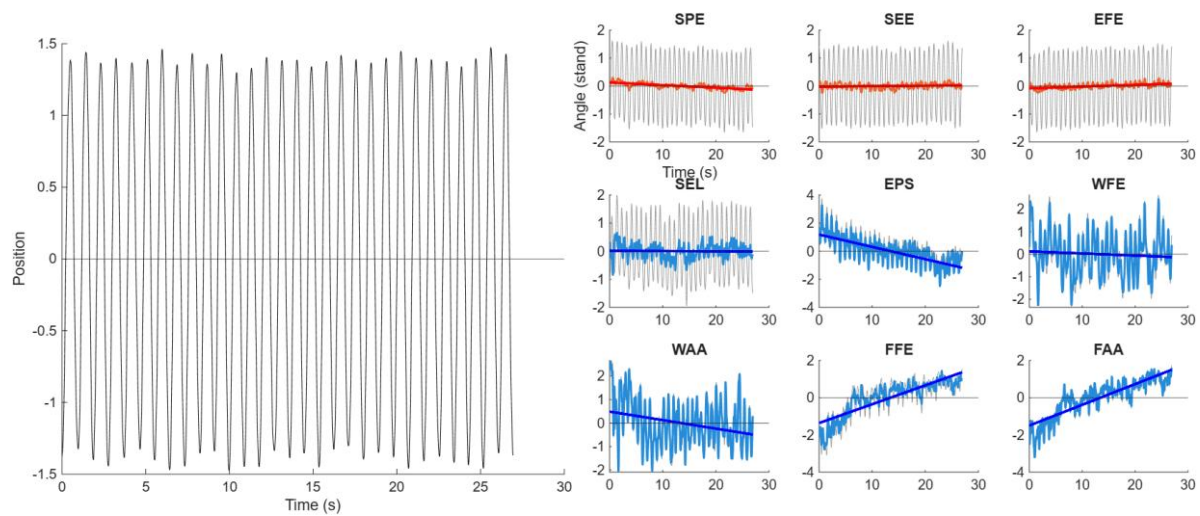


Figure 4

Detrended fluctuation analysis

To examine the spatiotemporal structure of the residual joint angle variability, we conducted a detrended fluctuation analysis (DFA) on all residual time series. DFA quantifies long range correlations in a time series by measuring how fluctuations scale across timescales, potentially signifying a fractal structure in the variability. This means it assesses whether patterns of variability repeat themselves across timescales. The output of DFA is a scaling exponent α , ranging roughly from 0 to 2. An α of 0.5 generally indicates randomness in the time series (white noise), whereas an α of 1.0 points at fractality (pink noise). A residual time series that expresses pink noise signifies that smaller, shorter fluctuations are embedded in larger, longer fluctuations. An α of 1.5 indicates non-stationarity or drift in the time series, also called Brownian noise. As shown in **Figure 5**, the DFA on the residual joint angle variability resulted in α -values between 1.0 and 1.5 in all angles, across all conditions. This indicated that the residual variability was not random, but structured. More particularly, it exhibited a mixture of fractality and drift, as already anticipated in **Fig. 4**. The indication of fractality was the strongest in the elbow flexion extension angle, while the suggestion of drift was the clearest for the wrist flexion extension angle.

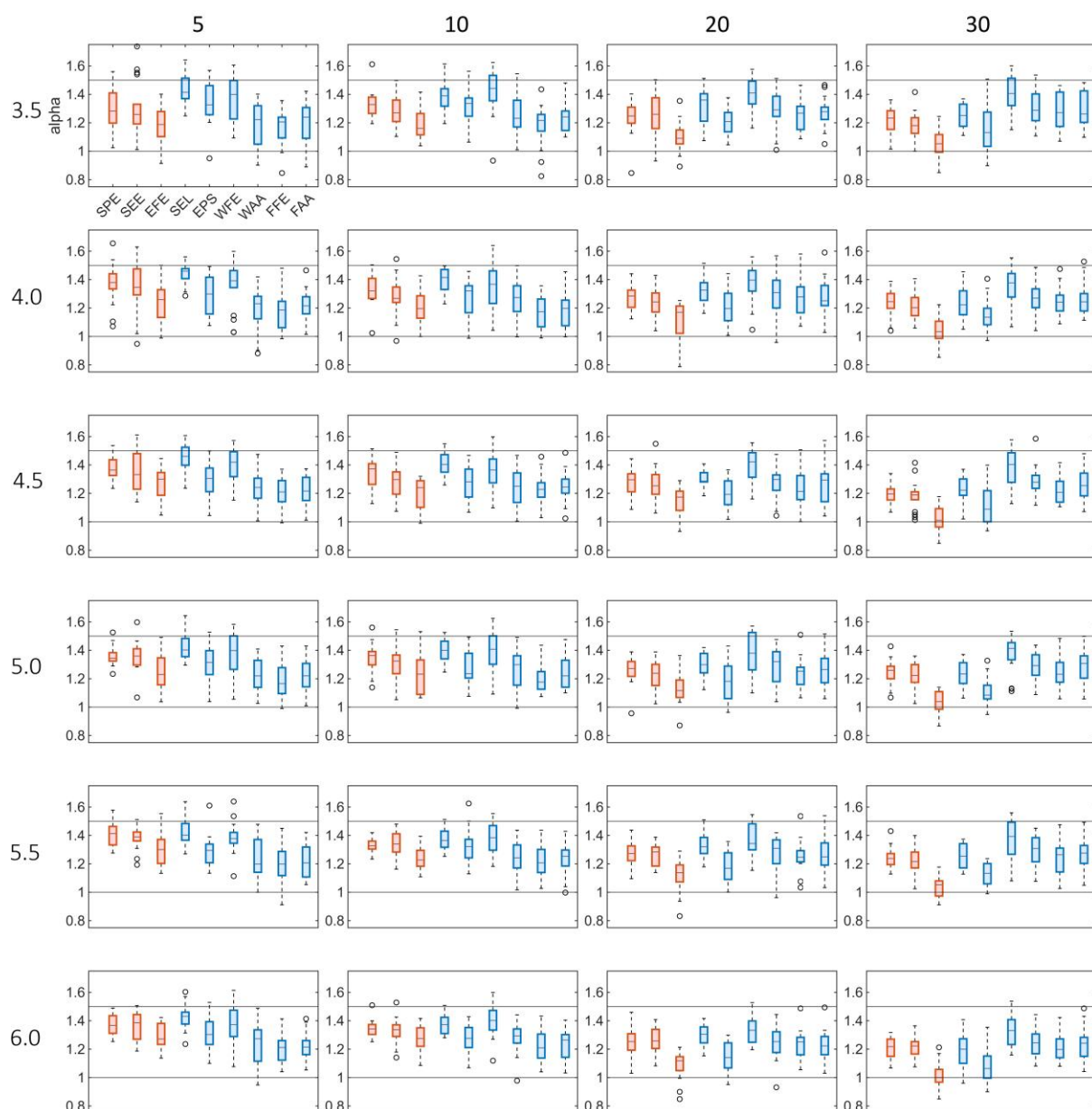


Figure 5

Drift analysis

To obtain a linear approximation of the drift, we fitted a first-order regression to the normalized angular time series, with time as a predictor and slope as our measure of drift. **Fig. 4** already showed these slope lines in the illustrated time series, demonstrating that the range and directionality of the drift (i.e., positive or negative drift) varied across the joint angles. In **Figure 6** we presented the distribution of the drift slopes across participants for all joint angles in all aiming conditions. As can be appreciated from this graph, all angles exhibited drift, even though the drift was larger for the non-synchronous angles (in blue). However, as shown in **Table 2**, the average range of motion (ROM) of these joint angles was also substantially lower than the ROM of the synchronous joint angles (in red). This means that the contribution of the non-synchronous joint angles to the upper extremity movement was smaller and the freedom to potentially ‘wander off from their trajectory’ without destabilizing the end-effector, larger.

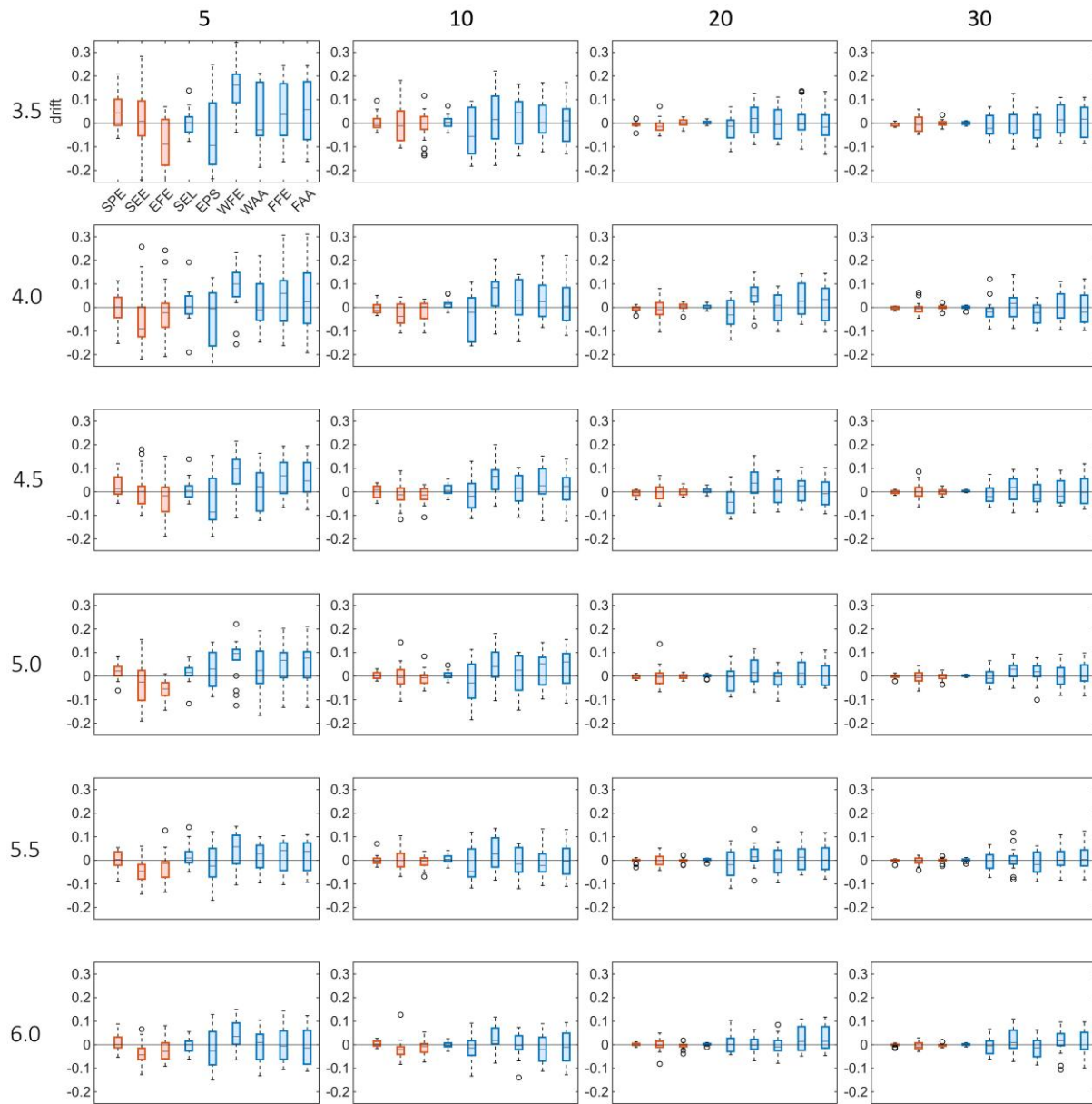


Figure 6

Table 2: The mean (standard deviation) range of motion of the joint angles in degrees, averaged per target size condition across all participants and all ID conditions.

Dist.	SPE	SEE	EFE	SEL	EPS	WFE	WAA	FFE	FAA
5 cm	9.1 (3.6)	10.8 (4.3)	10.0 (3.2)	4.7 (1.9)	6.0 (3.0)	5.8 (3.4)	3.6 (2.0)	2.1 (1.5)	3.2 (1.8)
10 cm	17.3 (5.1)	20.2 (6.0)	20.3 (4.3)	8.2 (3.2)	9.4 (3.8)	8.0 (4.3)	4.5 (1.8)	2.7 (2.2)	3.9 (2.2)
20 cm	32.7 (8.1)	37.8 (10.0)	40.0 (5.8)	13.0 (3.9)	15.0 (6.3)	11.7 (6.3)	5.9 (2.3)	3.5 (2.8)	4.8 (2.3)
30 cm	44.5 (10.2)	53.2 (13.4)	56.8 (7.7)	16.5 (4.2)	20.6 (8.7)	16.1 (8.6)	7.7 (3.0)	4.4 (3.8)	5.8 (2.8)

Projection length analysis

That joint angles exhibited drift while the end-effector dynamics remained stationary suggests that the drift occurred within the uncontrolled manifold [23] – the subspace in joint angle space that encompasses all joint angle configurations that stabilize the end-effector on a certain position. To investigate whether this was indeed the case, we projected the joint angle configurations of different cycles within a condition on the uncontrolled manifold as well as on the range space. The range space is a subspace in the joint angle space orthogonal to the manifold, including all joint angle configurations that lead to deviations of the end-effector from a certain position. We computed the projections for four instants of the cycle of motion: at the two moments of reversal at the physical target (indicated as 0 and 180 degrees of the cycle) and two moments mid-half-cycle (indicated as 90 and 270 degrees of the cycle). Using the joint angle configuration of the first cycle as reference, projection lengths for all four cycle moments increased when joint angle configurations were projected on the uncontrolled manifold, but not when projected onto the range space (**Figure 7**). This indicated that the drift observed in the joint angles was gradual and could be collectively described as drift along a subset of configurations that stabilizes end-effector behavior.

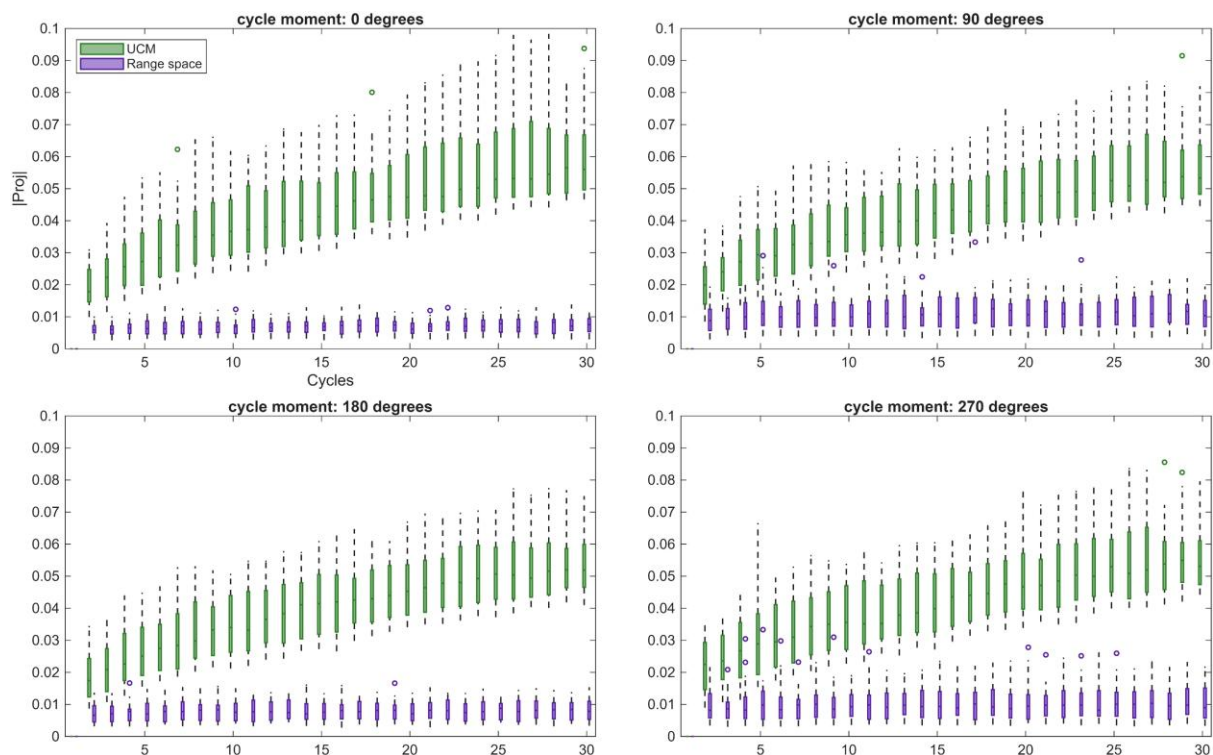


Figure 7.

Discussion

In this contribution, we investigated the coordination dynamics of upper extremity joint angles (as microscopic level) and the end-effector (as macroscopic level) during a reciprocal aiming paradigm. In the absence of well-established models for circular causality in multi-DOF effector systems, this investigation served to explore the process of self-organization in a redundant movement system. Our results revealed that the dynamical relationship between different joint angles and end-effector was not the same for all joint angles. Those joint angles with the largest range of motion synchronized with the end-effector. This synchronization suggests that dynamical coupling between the macro- and microscopic level of organization

is an important component of self-organization. At the same time, the joint angles exhibited variability that was not shared with the end-effector. We found indications of fractality and drift at the joint angle level, neither of which affected the stability of the end-effector dynamics. This tentatively suggests that the joint angles were not only coordinated to stabilize the end-effector, but also to keep the movement system flexible. In what follows, we will first discuss these results in more detail, before functionally interpreting them in perspective of the ecological-dynamical concept of synergy. We conclude the discussion by providing an outlook for future research into this topic.

In an earlier contribution exploiting the current data, Valk and colleagues [8] used a cross-recurrent quantification analysis to show that the shoulder plane of elevation angle, the shoulder exo-endorotation angle, the elbow flexion-extension angle and the end-effector were strongly coupled to one another. Here, we followed up on this finding with a continuous relative phase analysis. This analysis demonstrated that the coupling between joint angles and end-effector comprised stable in-phase or anti-phase relationships. This result indicates a high degree of synchronization between the macro- and microscopic level of organization, at least for some of the angles. Rather than preceding or following the end-effector, a subset of the joint angles moved *simultaneously* with the tip of the finger, effectively driving its macroscopic limit-cycle dynamics [12,25]. The other six joint angles, in particular those of the finger and wrist, were much more variable in their phase progression. Note that this relative phase variability persisted even when we controlled for drift and for variations in the amplitude. Using the tightened normalization procedures of de Poel et al. [29], we detrended and normalized each half cycle. However, the phase locking values of the wrist and finger angles remained low. Together, these results demonstrated substantial ‘residual’ variability in the angular time series, the structure of which we examined using a detrended fluctuation analysis (DFA).

Although all joint angles exhibited residual variability, the non-synchronous joint angles exhibited more fluctuations that were not shared with the end-effector than the synchronous angles, as expected. The DFA on the residual variability (the variability remaining in the joint angles after the shared, linear relation between end-effector and each of the joint angles was removed) revealed two intriguing properties. In the first place, it showed indications of fractality, meaning that smaller, shorter fluctuations were embedded in larger, longer fluctuations. This finding is interesting, because it suggests dynamical patterns of motion in the joint angles *on top of* the macroscopic dynamics present at the level of the end-effector. It shows that the variations in the joint angle motion on a slower timescale than the back-and-forth movement were not random, but (at least partially) deterministic. At the same time, we note that DFA results must be interpreted with caution, because the analysis cannot uniquely distinguish between fractality and nonstationary trends.

In motor control literature, long-range correlations have been linked to properties of flexibility and adaptability [29,34]. They were first found in locomotion patterns [35,36], but have also been demonstrated in other rhythmic movements such as rowing [37,38] and manual tapping [39,40]. The embedding of fluctuations of different time scales prevents a rhythm from becoming rigid, in the sense that every cycle of motion is the same. At the same time, fractality channels variability, since fluctuations are not random but correlated. As such, it has been presented as a “compromise between order and disorder”, keeping the

movement system adaptive, without degenerating into randomness or inflexibility [31,34]. Our results suggest that this may also be the case in reciprocal aiming, even though longer time series than available in the current experiment are needed to test this interpretation.

Secondly, the DFA revealed indications of Brownian noise, which suggests drift in the joint angle time series. Follow up (regression) analyses showed that most of the drift occurred *within* the uncontrolled manifold, suggesting covariation among the joint angles (see Valk et al. [8] for a broader discussion of the UCM). As to the underlying causes of this linear drift, we can only speculate. We considered that participants may have become fatigued over the course of a trial, causing them to slowly lower their elbow during the trial. Further examination of the upper extremity postures, however, did not reveal a systematic lowering of the elbow joint over cycles of motion. We also ruled out the possibility that the targets had shifted over time, because the sheet of paper was firmly attached to the table for the entire duration of a trial.

Together, this brought us to the tentative conclusion that the joint angle drift may have had a modest but functional role in the control of the aiming movement. Participants may have drifted towards a posture that was most comfortable for producing the aiming movement. Or they may have explored different ways to perform the task to warrant adaptability. Even though the drift effect was subtle and the preferred postures probably unique for every individual, our findings suggest that participants used the freedom to move along the UCM over the course of a trial so as to improve their aiming performance. This emphasizes that a singular focus on neither a relative phase nor a pattern of covariation is sufficient to capture the richness of coordination in redundant systems. Equally important is showing how participants move along the manifold, either gradually or in discrete steps [41]. Future research into more challenging perceptual-motor tasks should point out how the form and amount of drift in joint angles may be related to the end-effector control at a functional level.

What do these findings tell us about synergy, a concept that is as old as the discipline of movement science itself [1-3,7,9]? A notion of synergy often referred to in the literature is that of muscle synergy [50, 51, 52]. It commonly assumes co-activation of muscles that are organized into modules (called muscle synergies). This modular interpretation is difficult to reconcile with our finding of covariation between degrees of freedom, because co-activation and covariation can be opposite constructs. As such, our findings challenge the traditional perspective of muscle synergies. At the same time, we note that our analyses are confined to the level of the joint angles rather than the level of muscle activation patterns.

The modular interpretation of synergies stands in sharp contrast to the ecological–dynamical interpretation. Synergy, in this approach, is understood as an emergent structure of coordination with circular causality between its levels of organization [4,6]. Unfortunately, established models of circular causality in redundant systems like our movement system are currently lacking. As a stepping stone for the development of such models, our findings on reciprocal aiming suggest that a model of this kind should exhibit at least two coordinative features. To begin with, it should explain the *stabilization* of the kinematic outcome of the movement. Intriguingly, our results suggest that this stabilization may involve synchronization as well as covariation. Synchronization ‘drives’ the end-effector trajectory and occurs between the joint angles with the largest range of motion. Covariation ‘preserves’ the end-

effector trajectory and ensures that it maintains its dynamics between the targets. This interpretation aligns roughly with the understanding of synergy as promoted by Riley et al. [5]. In this view, synchronization serves to reduce the system's dimensionality, so that a macroscopic organization emerges from the many DOF in the movement apparatus. Covariation then preserves the integrity of that organization, so that it is robust against minor perturbations. This way, synchronization and covariation are two sides of the same coin – both bringing stability to movement.

Second, a model of circular causality in a redundant system should feature flexibility. Our analyses suggest that the joint angles of the upper extremity vary *systematically* from cycle to cycle during reciprocal aiming, even though aiming constitutes a highly stable movement. More specifically, the joint angles display slow drift and long range correlations across cycles of motion, neither of which affect the end-effector dynamics. This variability is notable, because it is not task-driven, cannot be attributed to fatigue and because it appears to be too structured for sensorimotor noise. It tentatively suggests that redundant, self-organizing systems are intrinsically variable to remain adaptive. Accordingly, we speculate that they fluctuate across cycles to warrant adaptability and that they drift over time to settle into a more comfortable or effective configuration.

Flexibility has been attributed to synergy before, most distinctly within the uncontrolled manifold approach [3]. But because the UCM was designed as a method to study the stabilization of a static performance variable, it is ill-posed to reveal the dynamical properties of flexibility in movement (see also Grover et al. [42]). As such, we undertook an explorative approach using non-linear time series analyses to reveal these properties. While our results suggest that flexibility manifests itself in fractality and unboundedness of the body's degrees of freedom, more research is needed to establish this relationship. Investigations of locomotion have already shown an empirical link between fractal scaling and adaptive capacity. For example, participants who exhibited 1/f noise in their gait patterns adapted more easily to a treadmill that induced gait asymmetries than those who did not [53]. Studies of this kind can clarify whether this relationship also applies to upper-extremity movements such as reciprocal aiming. Likewise, perturbation studies can test the here proposed hypothesis that fractal scaling in joint-angle variability provides the aiming movement with robustness against disturbances.

In conclusion, the present study has provided scaffolding for understanding and modeling circular causality in redundant movement systems. By examining concurrently the coordination dynamics of the joint angles and the end-effector during a reciprocal aiming task, we have shown that variability across cycles of motion is intrinsic to even the most stable forms of movement. This is, we propose, what self-organization in redundant movement systems may entail. The emergence of a coordinative structure –a synergy– that is stable and ready to withstand perturbations, but also variable and ready to adapt to the environment.

Method

Ethics statement

The experimental protocol was carried out in accordance with the Declaration of Helsinki and approved by the local ethics committee of the Department of Human Movement Sciences at

the University Medical Center Groningen. Before the start of the experiment, we informed participants about the goal of the study, after which all participants signed informed consent.

Participants

In total, we recruited twenty right-handed participants. Two participants were excluded because of failed data acquisition during the measurements. The remaining eighteen participants had an age of 20.8 ± 1.9 years ($M \pm SD$). They had no neurological conditions or other health issues and they all had normal or corrected-to-normal sight.

Experimental task

Participants made reciprocal aiming movements (i.e., back-and-forth in an anterior-posterior direction in front of them) between two targets, according to a continuous Fitts' paradigm [11,12] (**Figure 8**). We manipulated both target distance and index of difficulty (ID; a logarithmic function of target distance and target width), by varying the target width relative to the target distance. In total, the experiment comprised 24 unique conditions (trials), combining 4 target sizes (5, 10, 20 and 30 cm) with 6 ID's (3.5; 4.0; 4.5; 5.0; 5.5; 6.0). These conditions were presented to participants in a randomized order. In each trial, the targets were presented on a laminated sheet of paper (A3 size, portrait orientation) that was attached to the table in front of the participants. To avoid drifting of the end-effector in the frontal plane, all targets had a width of 1 cm in this plane. Participants made the pointing movements with a stylus that was attached to their index finger, leaving no trace to the paper. The stylus was attached such that it prohibited movement in the interphalangeal joints while allowing free movement of the metacarpophalangeal joint.

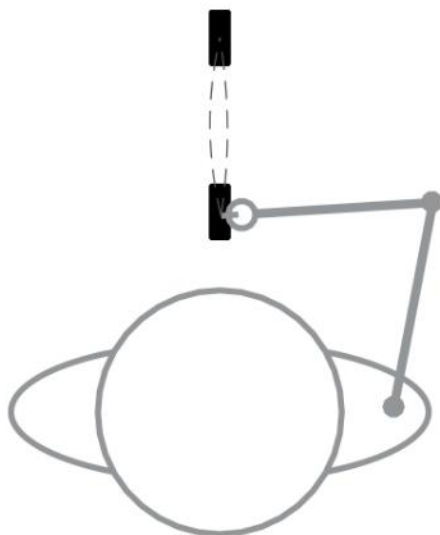


Figure 8.

Experimental procedure

In each trial, participants made forty reciprocal movements (cycles) between the targets. Participants always started a trial with the tip of the stylus in the middle of the target that was closest to them. They were instructed to take the same starting posture across conditions

– an instruction that was repeated before the start of every new trial. To check whether participants followed the instruction, we compared the standard deviation of the starting joint angle configurations across IDs within a target distance with the standard deviation of starting joint angle configurations recorded in previous studies in which a similar starting posture was ensured by means of an elbow placer [43,44]. This comparison showed that participants followed the instruction. During the trial, participants had to move the tip of the stylus as fast and accurately as possible. Experimenters observed whether this instruction was followed and motivated participants if necessary to move as fast as possible while adhering to accuracy demands. Such additional motivation was only provided to a few participants within the first trials, and only when participants very clearly moved the stylus too fast or too slow. Moreover, participants were instructed to always keep the tip of the stylus at the sheet of paper.

Data acquisition

To capture the participants' movements in three dimensions, we attached five rigid bodies to the right side of the body and one to the sternum. Each rigid body was triangular in shape and contained three light-emitting diodes. One rigid body, with a length of 4 cm, was attached to the stylus; four others were attached to segments of the participant's right arm and shoulder. Two rigid bodies, with a length of 6 cm, were attached to the sternum and the upper arm just below the insertion of the deltoid; three others, with a length of 4 cm, were attached to the dorsal side of the hand, the dorsal side of the upper arm just proximal of the ulnar and radial styloids, and the flat part of the acromion. We captured the movements of the LEDs using two Optotrak 3020 units (Waterloo, Ontario, Canada) that were synchronized and sampled at 100 Hz. Using a pointer device, we digitized eighteen bony landmarks and the tip of the stylus to relate the LED motion to the movement of the participant's arm. To prohibit movement of the trunk while allowing free movement of the shoulder joint, we gently strapped participants against the extended back of the chair using an elastic bandage.

We determined the end-effector trajectory from the motion of the three Optotrak LEDs attached to the stylus, using rigid body transformations. For further analyses, we exclusively used the forward-backward movement of the end-effector in the transversal plane. The trajectories of the nine joint angles we computed using the motion of the relevant Optotrak LEDs and the segment orientations derived from the digitized bony landmarks. Following ISB guidelines [45], we acquired the following joint angles: the shoulder plane of elevation (SPE), shoulder elevation (SEL), shoulder exorotation-endorotation (SEE), elbow flexion-extension (EFE), elbow pronation-supination (EPS), wrist flexion-extension (WFE), wrist abduction-adduction (WAA), finger flexion-extension (FFE), and the finger abduction-adduction (FAA).

Data analysis

The time series of the end-effector and the nine joint angles were filtered using a 4th order low-pass Butterworth filter with a cut off frequency of 5 Hz. To reconstruct the coordination dynamics at both levels, we created phase plane portraits of the end-effector and all joint angles. We normalized the (angular) position and (angular) velocity time series to their mean and standard deviation and plotted the latter against the former in to create phase planes.

Continuous relative phase analysis

To examine the synchronization between the joint angles and the end-effector, we computed the continuous relative phase between the corresponding time series. We cut all time series into half cycles, demarcated by the reversal points of the end-effector signal. Following the procedure of de Poel et al. [29], we then centered, detrended and normalized each half cycle relative to the standard deviation of that cycle, to avoid distortion of drift or amplitude. Subsequently, we transformed the half cycles of the end-effector and the joint angles to analytic signals using the Hilbert transform and used that to compute the phase angles in degrees [28]. After concatenating all half cycles, the continuous relative phase was then acquired by subtracting the phase angle of the joint angle from the end-effector. For interpretation, we plotted the CRP distributions per joint angle in polar histograms. Moreover, we computed the phase locking value per CRP time series to quantify the degree of synchronization. To define both in-phase and anti-phase as synchronization, we first doubled the CRP values, after which we took the complex phase vectors to represent each CRP value on the unit cycle. We then averaged all complex vectors on the unit cycle to compute the PLV. A PLV of 1 indicated maximum synchronization (in-phase/anti-phase), while a PLV of 0 indicated a random distribution of CRP values.

Detrended fluctuation analysis

To investigate the variability structure that deviated from the end-effector, we conducted a first order regression analysis on the normalized time series of the joint angles, with the normalized end-effector time series as a predictor. This resulted in a component of shared variability and a component of residual variability. This residual variability corresponded to the variation in the joint angle timeseries that was *not* shared with the end-effector, because the linear model was removed from the data. We analyzed this residual variability using a detrended fluctuation analysis (DFA), following standard procedures for periodic movement data [e.g. 37,38]. DFA quantifies long range correlations in a time series by measuring how fluctuations scale across timescales. As a first step, we integrated the residual variability time series. The total length (N) of a time series varied roughly between 2200 and 2900 samples, depending on the size of and distance between the targets. Subsequently, we divided each time series into windows of increasing length, from $n=10$ to $n = N/4$ with 20 scales in total. We then detrended each window and calculated its root mean square (RMS). As a last step, we derived a scaling exponent α from the slope of the log-log relationship between the RMS and the window size. The scaling exponent may range from 0 to 2. An α -value of 0.5 indicates the absence of correlations (white noise), $\alpha > 0.5$ indicates that large values are more likely to be followed by large values, and $\alpha < 0.5$ indicates that large values are more likely to be followed by small values and vice versa. Then, α -values around 1.0 and 1.5 correspond to pink noise and Brownian noise, respectively.

Drift analysis

To assess the drift in the time series of the joint angles, we first normalized the angular time series to z-scores by dividing them by their standard deviation. Thereafter, we fitted a first order regression on the normalized time series with time as a predictor, taking the slope of the regression model as our drift coefficient (z/sec). An inspection of the drift distribution revealed that both positive and negative drift values were present.

Projection length analysis

To compute the projection lengths onto the uncontrolled manifold of joint angle configurations, we used an adapted version of motor equivalence analysis [8,46-48]. To achieve these projections, first, a Jacobian matrix that describes how changes in joint angles relate to changes in end-effector position was constructed. Using linear regression procedures [49], for each condition, a Jacobian matrix was constructed for four cycle moments (at movement reversals at the target, and two mid-half-cycle moments during the movement in between two targets) using the joint configurations at those moments from all cycles within that condition. The null space of such a Jacobian matrix reflects the uncontrolled manifold containing all joint angle configurations that lead to a stable end-effector position, whereas the range space of this Jacobian matrix contains all joint angle configurations that lead to changes in end-effector position. To be able to project joint angle configurations onto both these spaces, joint deviation vectors (JDVs) were computed by subtracting the joint angle configuration of the first cycle (used as reference) from the joint angle configurations of subsequent cycles. Subsequently, these JDVs were projected onto both the uncontrolled manifold and the range space of the Jacobian to assess the evolution of projection lengths along these spaces across the different cycles of the various conditions (see Valk et al. [8] for details of the formulae used).

Data availability statement

The raw data supporting the conclusions of this article will be made available upon request.

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Figure legends

Figure 1: Normalized phase plane portraits of the end-effector (left) and the nine joint angles (right) for a representative participant (5) in an exemplary condition (distance 20 cm, ID 5.0). In a phase plane the (angular) velocity is plotted against the (angular) position. The graphs in red correspond to the joint angles that are synchronized with the end-effector, while the graphs in blue correspond to the angles that synchronized less with the end-effector.

Figure 2: Distribution of the continuous relative phase (in degrees) of all joint angles for a representative participant (5) in an exemplary condition (distance 20 cm, ID 5.0). The graphs in red correspond to the joint angles that are synchronized with the end-effector, while the graphs in blue correspond to the angles that synchronized less with that effector. For reasons of visibility, the x-axis varies in scale. Circles indicate the proportional frequency in steps of 0.25 on a scale of 0 to 1.00.

Figure 3: Distributions of the phase locking value (PLV) across all participants of the nine joint angle across all ID (horizontal axis) and target distance (vertical axis) conditions. Synchronized joint angles are represented in red, while non-synchronized joint angles are represented in blue.

Figure 4: Normalized time series (depicted in gray) of the end-effector (left) and the joint angles (right) for a representative participant (5) in an exemplary condition (distance 20 cm, ID 5.0). In the same plots, the residual variability for the joint angles that strongly synchronize with the end-effector (depicted in red) and that does not synchronize with the end-effector (depicted in blue) is presented. The trend line in each graph corresponds to the average linear drift in the joint angle over time.

Figure 5: Distributions of the alpha-value (DFA) across participants of the nine joint angle across all ID (horizontal axis) and target distance (vertical axis) conditions. Synchronized joint angles are represented in red, while non-synchronized joint angles are represented in blue. Lines indicate alpha values of 1.0 and 1.5, between which most of the data is located. See text for interpretation.

Figure 6: Distributions of the drift coefficients of the nine joint angle across all ID (horizontal axis) and target distance (vertical axis) conditions. Synchronized joint angles are represented in red, while non-synchronized joint angles are represented in blue.

Figure 7: Projection lengths of joint angle configurations onto both the uncontrolled manifold (in green) and the range space (orthogonal to this manifold, in purple), for four movements during the movement cycle between the two targets. 0 and 180 degrees indicate the moment of movement reversal while 90 and 270 degrees indicate the mid-half-cycle moments. Boxplots display the range of projection lengths across all conditions at cycle n, using mean projection length across participants per condition as input data for these boxplots.

Figure 8: Experimental set-up, as seen from above. Note that the distance between targets and width of the targets (in the forward-backward direction with respect to the participants heading) was adjusted across conditions. Dashed lines illustrate the end-effector trajectory for one cycle of pointing movements. Figure copied with permission from Valk et al [8].

Table 1: The mean (standard deviation) range of motion of the joint angles in degrees, averaged per target size condition across all participants and all ID conditions.

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Author contributions

M.S.J.H., T.A.V. and R.M.B. conceived this study. T.A.V. and R.M.B. designed the original experiment. T.A.V. conducted the experiment. M.S.J.H and T.A.V. wrote the programs to run the data analysis. M.S.J.H., T.A.V., R.M.B. interpreted the results. M.S.J.H., T.A.V. and R.M.B. wrote the manuscript and approved the final version of the paper.

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Additional information

Competing interests

We declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.